



The duck-billed, egg-laying, web-footed platypus, first described by British zoologist George Shaw, is amphibious—capable of living both on land and in water.

A year later, a specially made platinum cylinder with a mass equal to 1.000025 liters of water, was legally declared the **official prototype of the kilogram**. The modern kilogram is derived from this prototype.

In July 1799, **Pierre Bouchard** (1772–1832), an officer in Napoleon's French army, found a black granite stone at Rosetta, Egypt. The stone carried the **same text in three scripts**—hieroglyphs, Egyptian demotic script, and ancient Greek. This provided the key to translating hieroglyphs. The **Rosetta Stone**, was among the items captured by the British in 1801.

In the same year, French astronomer **Pierre-Simon Laplace** (1749–1827) published the first part of his five-volume *Traité de Mécanique Céleste* (*Treatise on Celestial Mechanics*). Described as “an encyclopaedia of calculations about the six known planets and their satellites, their shapes, and tides in Earth's oceans,” it proved that the Solar System is stable on timescales that are relevant to humankind.

In 1799, physician **Thomas Beddoes** (1760–1808), established the **Pneumatic Institution** in Bristol, UK, to research the medical implications of newly discovered gases. It was here that British chemist Humphry Davy served his apprenticeship.

THIS YEAR, ITALIAN SCIENTIST ALESSANDRO VOLTA (1745–1827) wrote to the Royal Society, London, to report the battery he had made, which became known as a **voltaic pile**. His invention was actually a “**wet cell**” battery. In 1799, Volta had found that stacking disks of zinc, copper, and cardboard soaked in brine in multiple layers could generate an electric current. Adding more disks increased the amount of electricity generated.

The year also produced a puzzle for scientists, when British zoologist **George Shaw** (1751–1813) published the first scientific description of a **platypus** (*Ornithorhynchus anatinus* Shaw), based on preserved samples and sketches sent from Australia. He considered the creature so

bizarre that, at first, he thought it might be a hoax perpetrated by an accomplished taxidermist.

Another surprise came when British-German astronomer **William Herschel** studied the heat associated with the **different colors of sunlight, as split into a spectrum by a prism**. Herschel placed thermometers in different parts of the spectrum and found that the temperatures of the colors increased from the violet part of the spectrum to the red. He then placed a thermometer just beyond the red and observed that this had the highest temperature of all. This invisible radiation is now called **infrared** (see pp.234–35).

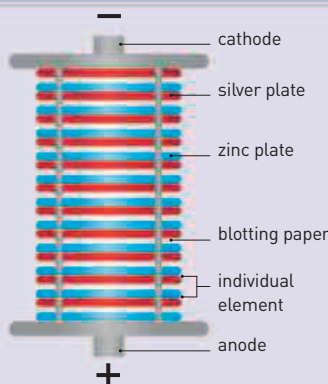
In industry, an important step toward the era of precision engineering came when British

engineer and inventor **Henry Maudslay** (1771–1831) developed the first practical **screw-cutting lathe**—a machine for accurately cutting screw threads.

In 1800, the **Royal Institution of Great Britain**—based in London—was granted its charter. Its aim was to provide a more popular forum for science than the established Royal Society. Henry Cavendish and Benjamin Thompson were instrumental in founding this institution, and Humphry Davy soon became its director. It became a **major research center** and a center for the popularizing of science throughout the 19th and into the 20th century.

HOW BATTERIES WORK

Batteries contain one or more electrochemical cells. Chemical reactions encourage electrons to move to the cathode end of the battery, building up negative charge there. Positive charge collects at the anode. When the cathode and anode are connected by an external wire, the electrons flow along the wire to produce an electric current. This is a simple “wet cell” battery.



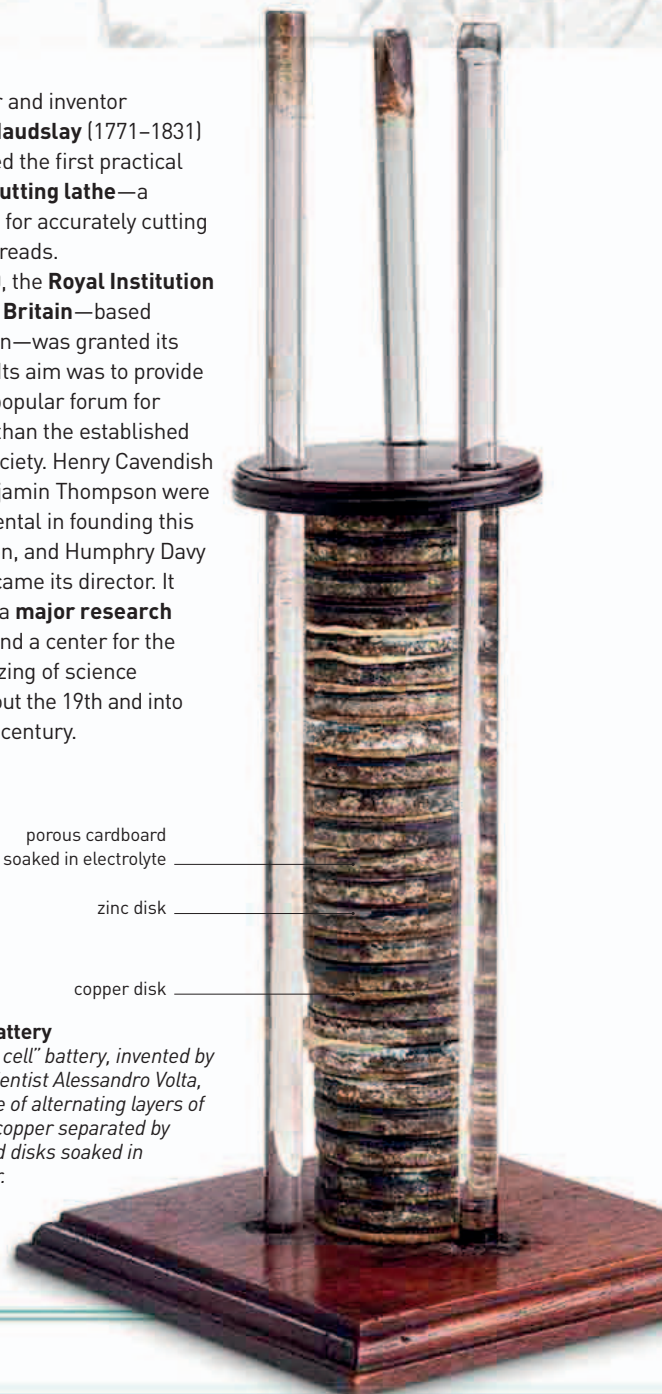
porous cardboard soaked in electrolyte

zinc disk

copper disk

Volta's battery

This “wet cell” battery, invented by Italian scientist Alessandro Volta, was made of alternating layers of zinc and copper separated by cardboard disks soaked in salt water.



1799 Thomas Beddoes establishes the **Pneumatic Institution** in Bristol

1799 Humphry Davy argues that **heat** is not a fluid called caloric but that it is “**motion**”

1799 Prototype kilogram is manufactured

Royal Institution of Great Britain is granted its charter

William Nicholson and Anthony Carlisle discover **electrolysis**

Alessandro Volta describes the **voltaic cell** in a letter to the Royal Society

Yeast is used as a fermenting agent in **beer** for the first time

George Shaw describes the **platypus**

William Herschel discovers **infrared radiation**

Henry Maudslay develops the first practical **screw-cutting lathe**